



DELHI PUBLIC SCHOOL NEWTOWN
SESSION: 2023-24
MONDAY TEST

CLASS : X
SUBJECT : CHEMISTRY

FULL MARKS : 40
DATE: 06.11.2023

General Instructions:

- The paper consists of three printed pages.
- Answers should be to the point.
- Question numbers should be copied carefully while answering the questions.

SECTION A
(Attempt all questions)

Question: 1

Choose the correct answer from the options given below: [8]

- (i) Which of the following pair of compounds have same empirical formula?
- Ethane and Ethyne
 - Ethene and Ethyne
 - Acetylene and Ethylene
 - Acetylene and Benzene
- (ii) The temperature at which acetic acid freezes to form ice like mass:
- 19.5°C
 - 15.5°C
 - 16.5°C
 - 18.5°C
- (iii) Which of the following is the correct value of Avogadro's number:
- 60.23×10^{23}
 - 60.23×10^{22}
 - 60.23×10^{25}
 - 60.23×10^{21}
- (iv) Which of the following contains maximum number of molecules at STP?
- 44 g of CO₂
 - 32 g of SO₂
 - 18 g of NO₂
 - 35.5 g of Cl₂
- (v) Which of the following is an example of chain isomerism:
- Ethane and Ethene
 - Butene and 2-Butene
 - Propanal and Propanol
 - 2-methylbutane and 2,2-dimethylpropane

- (vi) The number of moles present in 5.6 L of ethane is:
 A. 0.75
 B. 0.50
 C. 0.25
 D. 0.01
- (vii) The reagent which can be used to distinguish between ethane and ethene is:
 A. Ammoniacal cuprous chloride.
 B. Bromine water
 C. Cold dilute alkaline potassium permanganate solution
 D. Ammoniacal silver nitrate
- (viii) The gas released when a piece of sodium is added to ethanol is:
 A. Hydrogen
 B. Oxygen
 C. Carbon dioxide
 D. Carbon monoxide

Question: 2

Name the following: [5]

- (i) An amphoteric oxide which acts as a dehydrating agent at a temperature of 350°C.
- (ii) The metal used as a catalyst for hydrogenation reaction..
- (iii) The self-linking property of carbon atoms to form long chains via covalent bond formation.
- (iv) The amount of substance which contains the same number of units as the number of atoms in 12.000 g of carbon – 12.
- (v) An unsaturated hydrocarbon used for welding purpose.

Question: 3

What would you observe when: [2]

- (i) Ethyne is bubbled through bromine water.
- (ii) Ethane is burnt in excess supply of air.

SECTION B

(Answer any 5 questions from this section)

Question: 4

Give reasons for the following: [5]

- (i) Ethane does not undergo addition reaction.
- (ii) Ethyne is more reactive than ethene.
- (iii) 90% of the compounds in nature are organic.
- (iv) Alkanes are considered to be excellent fuels..
- (v) Burning of alkanes in limited supply of air is dangerous..

Question: 5

Solve the following numericals: [5]

- (i) $P + 5HNO_3 \text{ (conc)} \rightarrow H_3PO_4 + 5NO_2 + H_2O$. If 9.3 g of phosphorus is used in the reaction given, calculate:
 - a) Number of moles of nitric acid taken.
 - b) The mass of phosphoric acid formed

- c) The volume of NO_2 produced at STP.
 [P = 31, O = 16, N = 14, H = 1]
- (ii) Calculate the percentage of water of crystallisation in $\text{MgSO}_4 \cdot 6\text{H}_2\text{O}$. [Mg = 24, S = 32, O = 16, H = 1]

Question: 6

Write balanced chemical equation for the following:

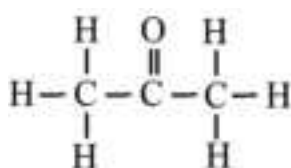
[5]

- (i) Decarboxylation reaction.
- (ii) Dehydrohalogenation reaction.
- (iii) Esterification reaction.
- (iv) Halogenation reaction.
- (v) Dehydration reaction.

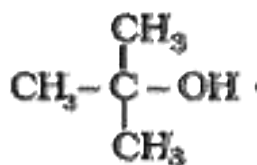
Question: 7

Write the IUPAC name of the following:

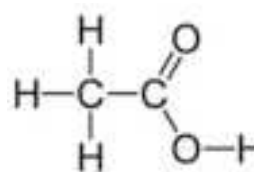
[5]



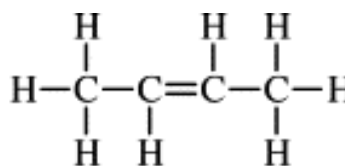
(i)



(ii)



(iii)



(v)



(iv)

Question: 8

Draw the structure of the following:

[5]

- (i) Acetaldehyde
- (ii) Neo pentane
- (iii) 2-Chlorobutanoic acid
- (iv) Formic acid
- (v) Chloroform

Question: 9

[5]

- (a) 3.5 g of nitrogen combines with 2 g of oxygen to form an oxide of nitrogen. What is the empirical formula of the compound?
 [N = 14, O = 16]
- (b) 2500 cc of oxygen was burnt with 600 cc of ethane. Calculate the volume of unused oxygen and volume of carbon dioxide formed. The equation for the reaction is: $\text{C}_2\text{H}_6(g) + 7\text{O}_2(g) \rightarrow 4\text{CO}_2(g) + 6\text{H}_2\text{O}(g)$